UNIT-1:
Power BI Foundations & Data Modeling

1 Power BI Desktop, Power BI Service & Power BI Workflow

What is Power BI?

Power BI is a **business intelligence (BI) and data visualization tool by Microsoft** used to convert raw data into meaningful insights using interactive reports and dashboards.

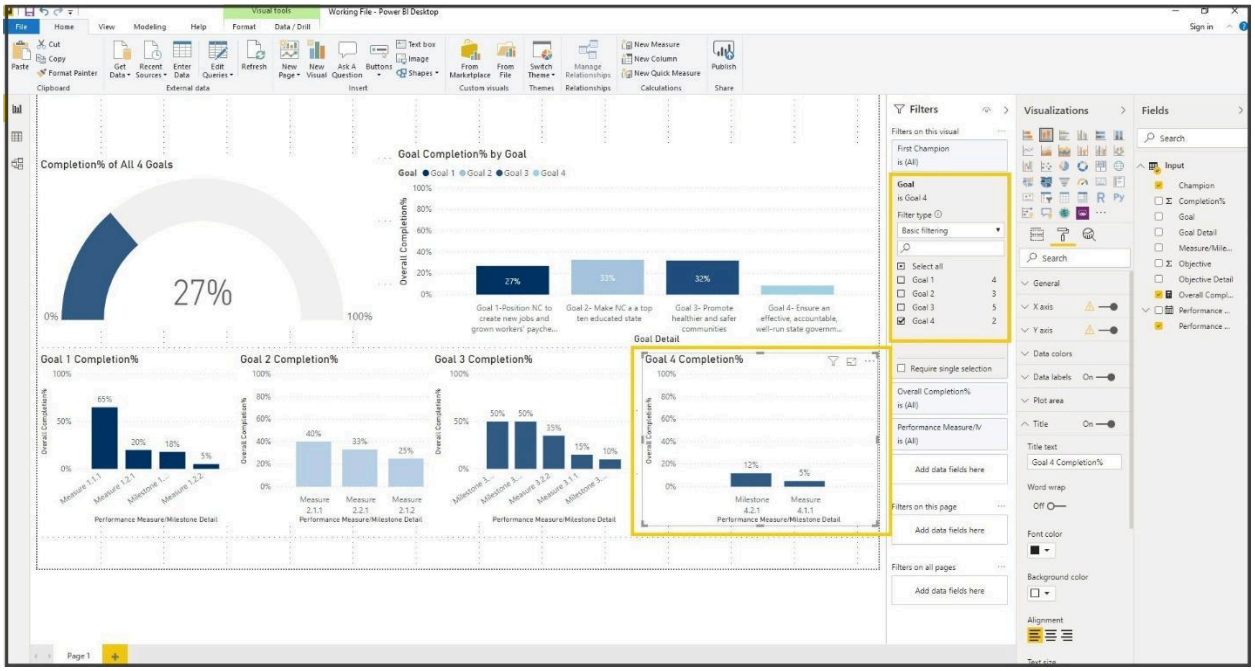
It is mainly used by **data analysts, business analysts, and decision-makers**.

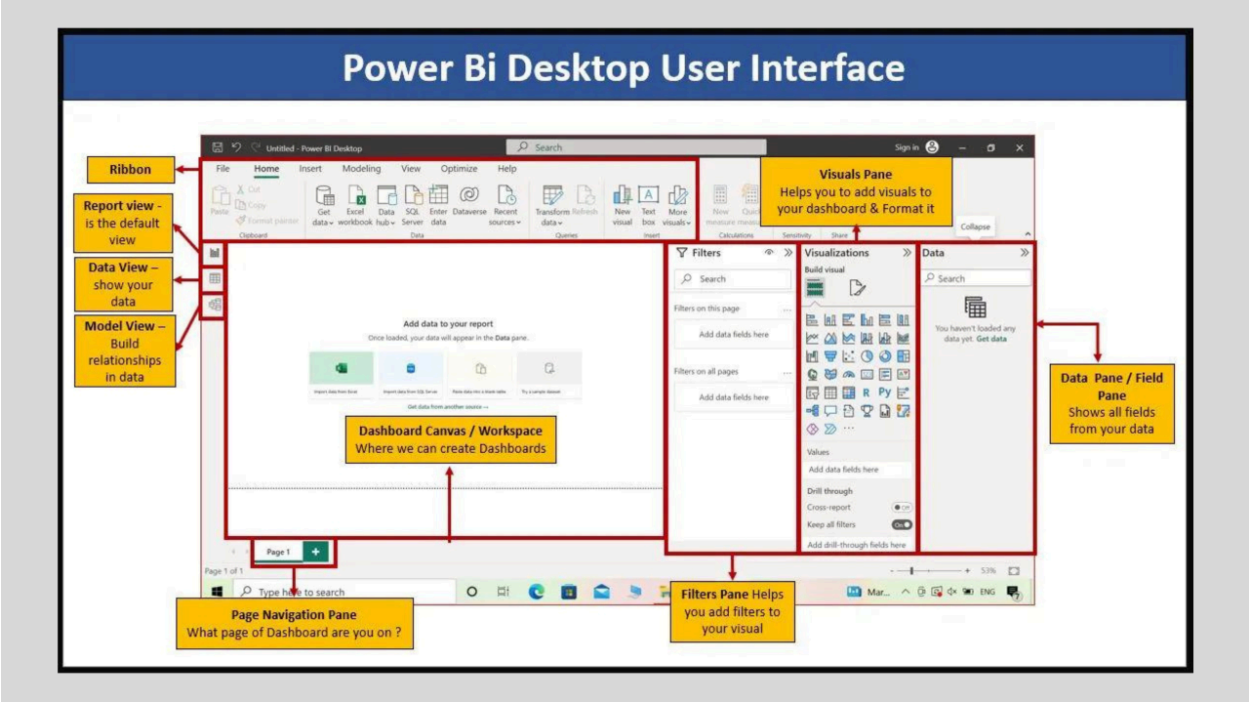
Power BI Desktop

Power BI Desktop is a **Windows-based application** used for: - Connecting to data sources - Cleaning and transforming data (Power Query) - Creating data models and relationships - Writing DAX calculations - Designing reports and visuals

Important point: All development work is done in Power BI Desktop.

Example: An analyst connects an Excel sales file, cleans the data, creates relationships, and builds charts using Power BI Desktop.



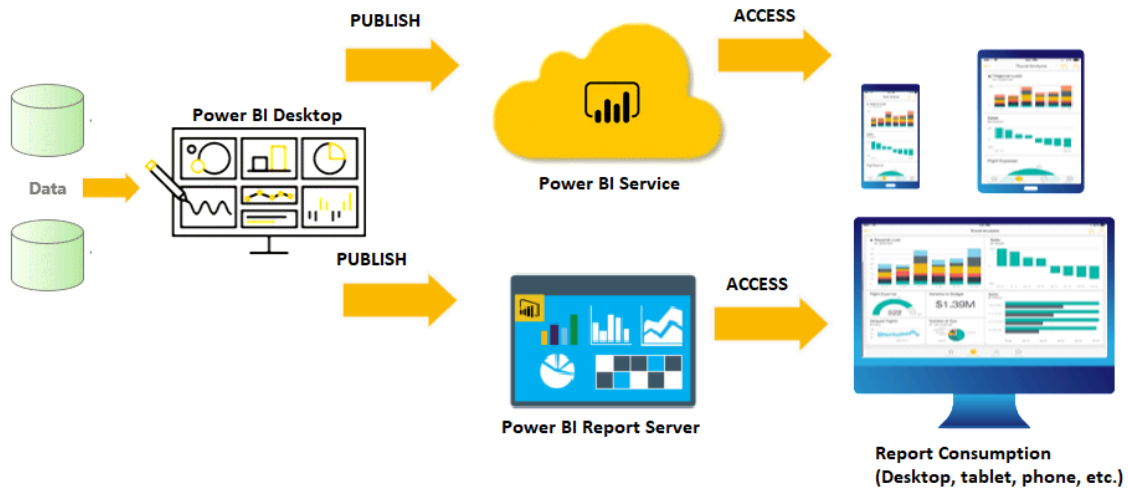


Power BI Service

Power BI Service is a **cloud-based platform (powerbi.com)** used for: - Publishing reports from Desktop - Sharing reports with users - Creating dashboards - Scheduling data refresh - Managing user access and security

Important point: End users mostly interact with Power BI Service.

Example: A manager opens a sales dashboard on Power BI Service to view daily performance.



Power BI Workflow

The standard workflow in Power BI follows these steps:

1. **Get Data** – Connect to data sources
2. **Transform Data** – Clean and shape data using Power Query
3. **Model Data** – Create relationships and measures
4. **Build Reports** – Create visuals and dashboards
5. **Publish** – Upload to Power BI Service
6. **Share & Refresh** – Share reports and schedule refresh

2 Data Sources in Power BI (Excel, CSV, SQL)

What are Data Sources?

A data source is the **location from which Power BI fetches data**.

Power BI supports **hundreds of data sources**.

File-Based Data Sources

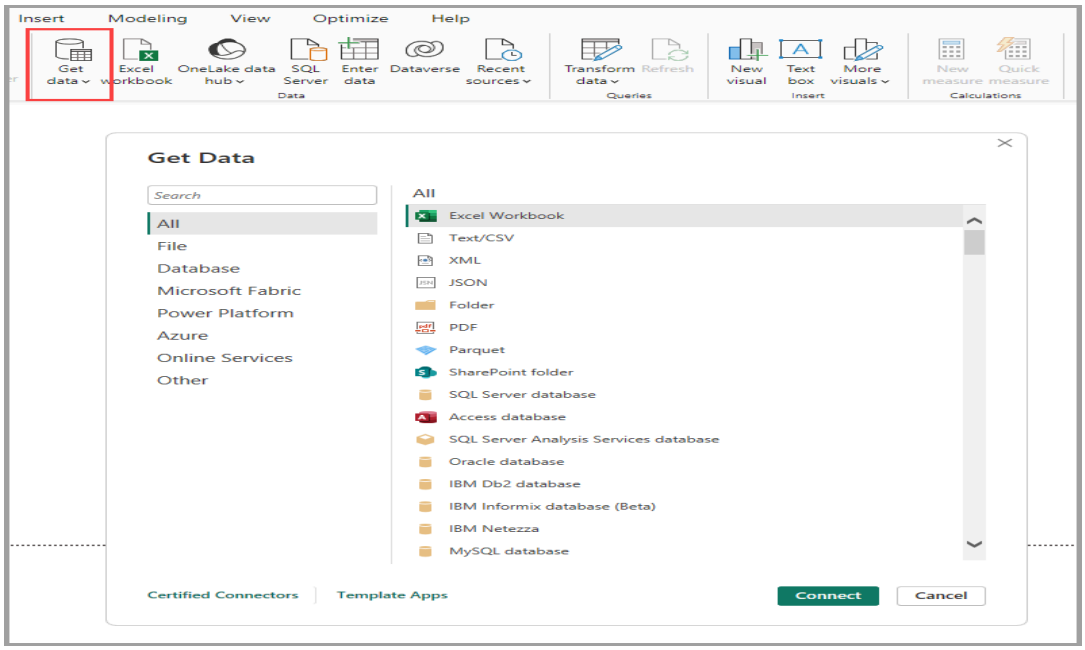
- Excel (.xlsx, .xls)
- CSV / Text files
- XML / JSON

Example: An MCA student analyzes student marks stored in an Excel file.

Database Data Sources

- SQL Server
- MySQL
- PostgreSQL
- Oracle Database
- Azure SQL Database

Example: An e-commerce company stores sales data in SQL Server, which is connected to Power BI.



3 Import Mode vs DirectQuery Mode

Import Mode

In Import mode, data is **loaded into Power BI memory**.

- Data is **copied into Power BI**
- Reports are **very fast**
- Works **offline** after import
- Data updates only when you **refresh**
- Best for **small to medium data**

Characteristics: - Very fast performance - Full Power Query and DAX features - Requires manual or scheduled refresh

Example: Monthly sales data imported into Power BI for analysis.

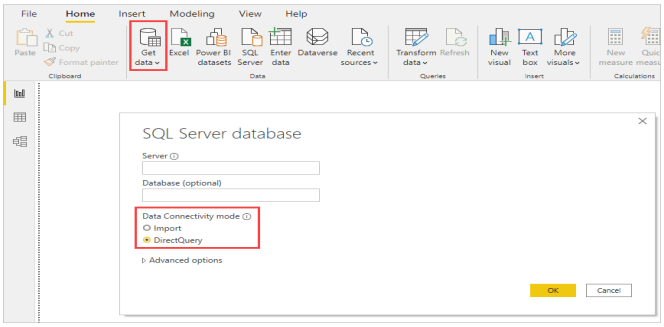
DirectQuery Mode

In DirectQuery mode, data **remains in the source database**.

- Data stays in the **original database**
- Power BI **queries live data**
- Needs **constant internet/database connection**
- Slower than Import (depends on database speed)
- Best for **very large or real-time data**

Characteristics: - Real-time data access - Slower compared to Import - Limited transformations

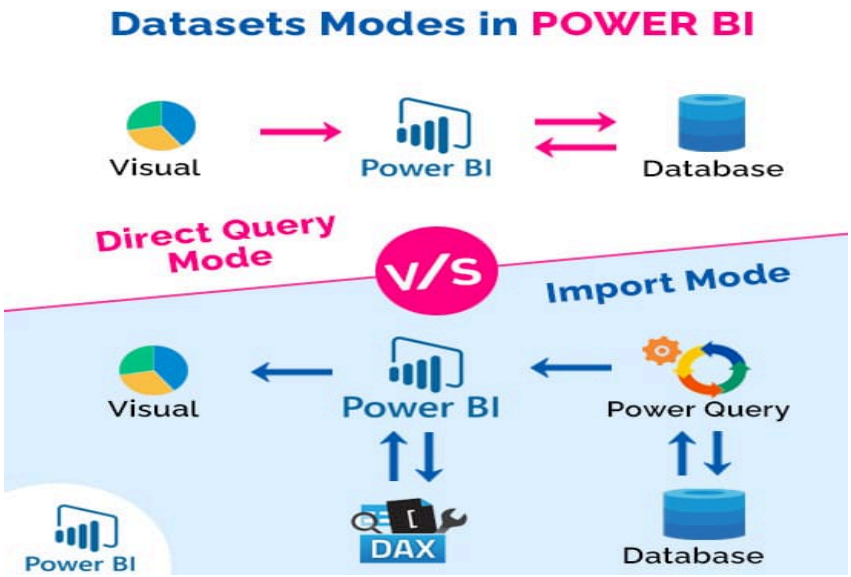
Example: Live inventory tracking system connected directly to SQL Server.



Import vs DirectQuery

Feature	Import	DirectQuery
Speed	Fast	Medium/Slow
Data Storage	Power BI	Source DB
Refresh	Required	Not required

Import vs DirectQuery comparison diagram



4 Relationships & Cardinality

What is a Relationship?

A relationship defines **how tables are connected** using a common column.

Example: CustomerID in Sales table linked to CustomerID in Customer table.

Cardinality Types

One-to-Many (1: n)

- **Most common**
- One row in **Products** → many rows in **Sales**
- Example:
  Customers (1) → Orders (many)

Many-to-One (n:1)

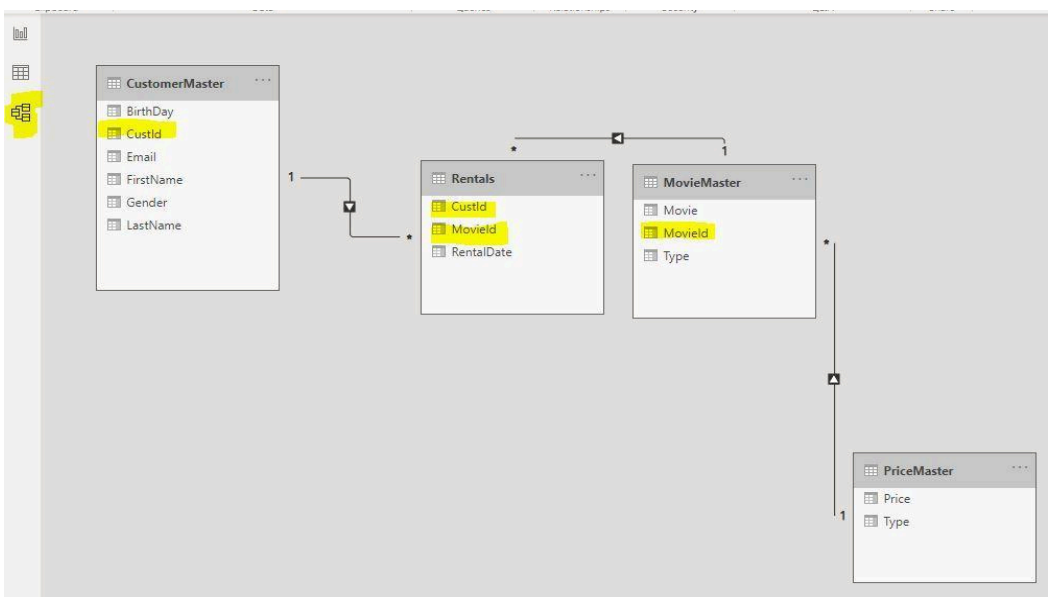
- Same as above, just reversed
- Orders (many) → Customers (1)

One-to-One (1:1)

- One row matches one row
- Rare
- Example:
  Employee ↔ EmployeeDetails

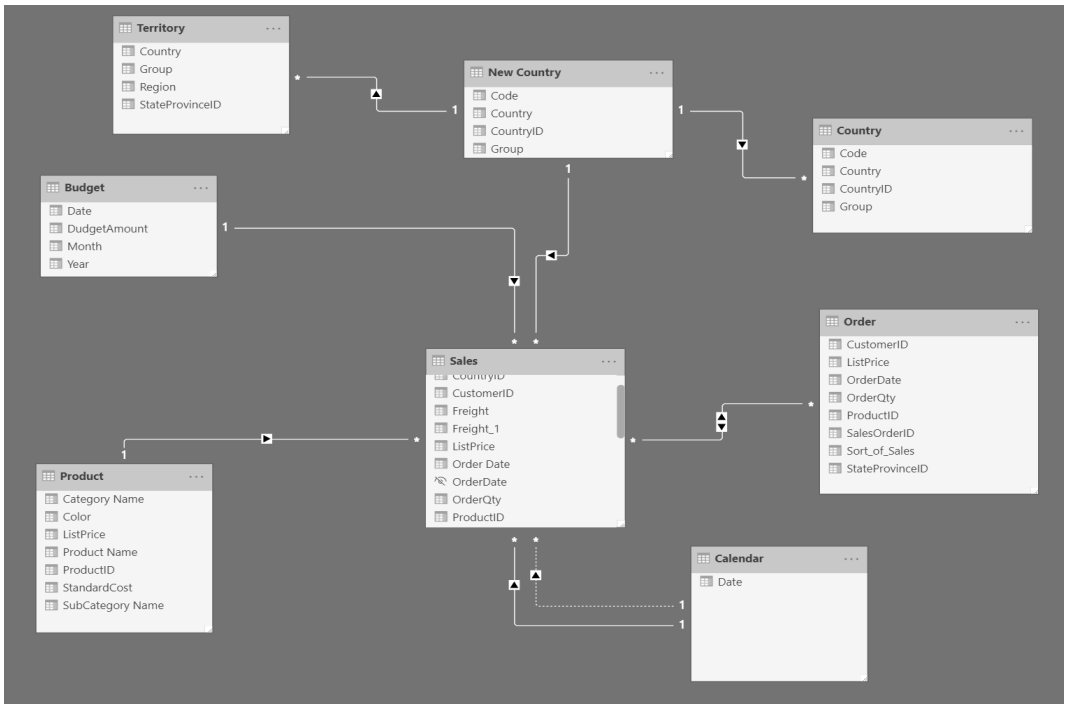
Many-to-Many (n:n)

- Many rows match many rows
- Used when **no unique key** exists
- Example:
  Students ↔ Courses



Notes: Arrow Direction

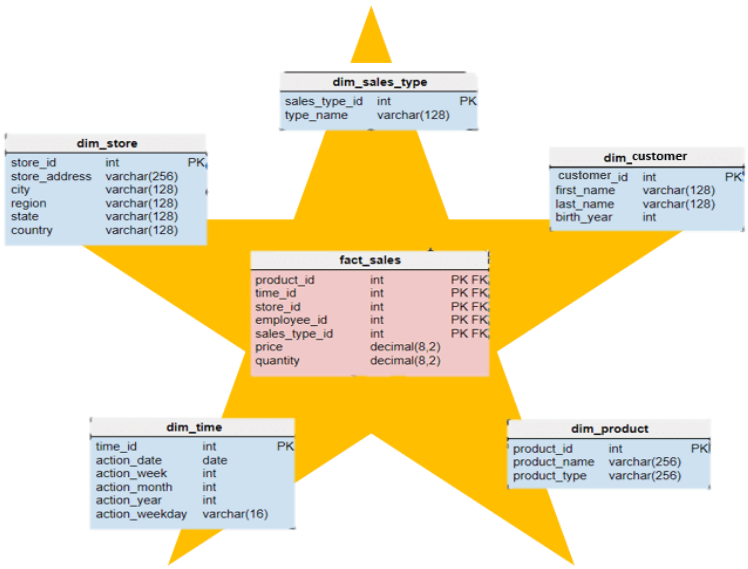
- Arrow shows **filter direction**
- Filters flow from **1 side → n side**
- When you filter **Customers**, it filters **Sales**
- This is called **Single Direction** (recommended)



5 Star Schema (Fact & Dimension Tables)

What is Star Schema?

Star schema is a **data modeling technique** where: - A central **Fact table** is connected to multiple **Dimension tables**




```
        Customers
            |
Products — Sales — Date
            |
        Stores
```

Fact Table

- Contains **numeric/measurable data** (Sales, Quantity, Profit)
- Very **large** table
- Has **foreign keys** to dimension tables

Contains: - Numeric measures (Sales, Quantity, Revenue) - Foreign keys linking to dimensions

Example: Sales_Fact(SaleID, ProductID, DateID, Amount)

Dimension Table

• Contains **descriptive data** (Customer, Product, Date)
• Smaller tables
• Used for **filtering and grouping**

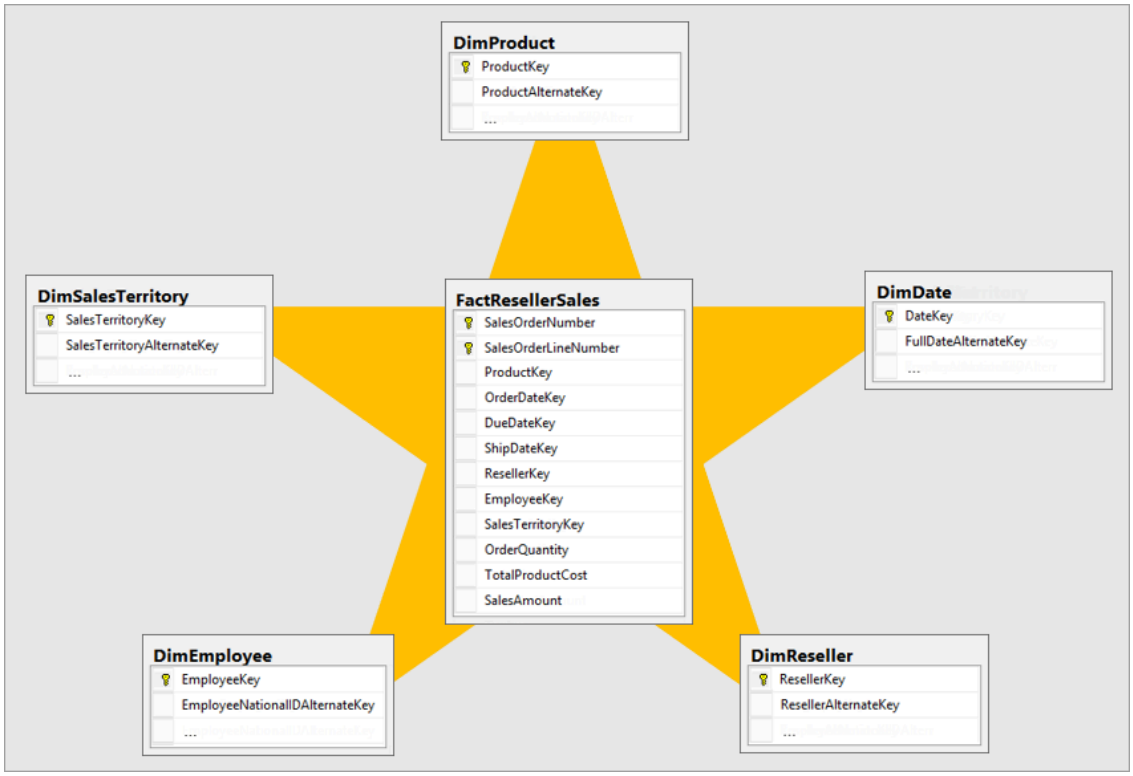
Contains: - Descriptive information - Used for filtering and grouping

Example: Product_Dim(ProductID, ProductName, Category)

Why is the Star Schema Important?

- Better performance
- Easy DAX calculations
- Clean and scalable model

🖼️ *Star schema diagram (Fact center + Dimensions)*



📝 Practice Questions (Unit-1)

Short Answer

1. Define Power BI.
2. What is Power BI Desktop?
3. What is Power BI Service?

Descriptive

4. Explain Power BI workflow with diagrams.
5. Differentiate Import and DirectQuery modes.
6. Explain relationships and cardinality.
7. Describe a star schema with an example.